Cardiff Metropolitan University | Prifysgol Metropolitan Caerdydd

School of Technologies

Assessment Brief

Module Code	Module Title
CMP7005	Programming for Data Analysis

Academic Year	Semester
2025-2026	2

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Assessment Details

Assessment title	Abr.	Weighting

From Data to Application Development	PRES1	30%

Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.

Task/Assessment brief:

- **Assessment Requirements**

Your task is to prepare a **10-15-minute presentation using your PRAC1 work**. You need to cover the following points:

a. **Motivation for Investigating the Dataset:**
   - Describe your motivation for analysing the dataset provided.
   - Which aspects of the data or domain are important to explore, and why?
   - What outcomes or deeper insights do you hope your analysis will reveal?

b. **Data handling:** Clarity in describing the data import process, including any challenges faced and how they were addressed.

c. **Critical Evaluation:** Critically assess the methodologies (e.g., data cleaning techniques, EDA approaches, machine learning models) you used for your analysis. How do these choices contribute to your understanding of the data and the domain?

d. **Key Findings:** Discuss the insights gained from analysis results. Use clear and informative visualizations to effectively communicate the key findings.

e. **Application Development:** Demonstrate the working software or platform you developed as part of the project. Show how it integrates with your project.

f. **Critical Reflection:**
   - ✓ Provide actionable recommendations based on your findings and the development of your software platform.
   - ✓ Discuss any challenges or limitations you encountered, such as data quality issues or model constraints, application development, and
   - ✓ Identify areas for improvement and suggest how these could be addressed in the future.

g. **Version Control:** Reflect on your use of version control throughout the project. How did it help you manage and organize your work?

📝 **Note: For PRES1, you are required to submit a shareable link to your recorded presentation. The recording may be uploaded to YouTube, Microsoft Stream, or any other appropriate platform. Please ensure that the following requirements are met:**

- ✓ The link is accessible and working properly.
- ✓ Appropriate sharing permissions are enabled (e.g., anyone with the link can view).
- ✓ Your recording has clear audio — please check for voice clarity and any other relevant factors before submitting the link.

Word count (or equivalent):	1200 words

This a reflection of the effort required for the assessment. Word counts will normally include any text, tables, calculations, figures, subtitles and citations. Reference lists and contents of appendices are excluded from the word count. Contents of appendices are not usually considered when determining your final assessment grade.

Academic or technical terms explained:

EDA- Process of analysing and investigating data sets and summarizing their main characteristics, often employing data visualization methods

.mp4, .mov, .avi – Common video file formats

Comprehension – Understanding how to access datasets and perform modifications for data analysis.

Analysis - Evaluate commonly used libraries used for data analysis tasks to develop solutions on a real-world dataset.

Synthesis- Integrating different programming concepts and techniques to develop functional software.

Evaluation- Assessing the effectiveness of data analysis for effective decision-making.

Application- Implement functional software and apply version control in practice.

Artificial Intelligence Models – Guidance for this assessment:

Artificial Intelligence (AI) models can be a powerful tool to support your learning. The University has provided some resources to support you in its appropriate usage:

- Library Services AI Hub
- Student Guide to AI and Assessment
- Code of Conduct for Students on the use of AI
- Cite Them Right resource on citing materials relating to AI (if permitted)

As per the academic regulations (Academic Handbook Ah1_08), in all cases you must submit work that is your own, acknowledging any part of it that has been informed by another source – including that which is AI generated. Upon submission of work, you will be asked to confirm the following statement:

> *I confirm that this assignment is my own work, except where I have acknowledged the use of works from other sources, including the use of any artificial intelligence (AI) tools, in accordance with what is allowable as described in the assessment brief.*

Please note the following:

- AI should not be used as a substitute for your own knowledge, and you should never include any material that you do not understand and could not explain if asked.
- Not being able to explain your work when asked is likely to be a key factor when considering cases of academic misconduct related to AI.

The following information provides specific guidance for this assessment about what level of AI use is appropriate for this assessment. Remember that in all cases you must submit work that is your own, acknowledging any part of it that has been provided by another source.

NO USE OF GENERATIVE AI EXPECTED • Your assignment should be produced using information sourced by you from your learning materials and academic sources and cited appropriately. • AI tools for checking spelling, grammar and referencing may be used.	
AI ACKNOWLEDGED • You can use AI tools to learn about your topic, as part of your study, or in preparing initial guidance on assignments (e.g. headline structure, suggestions for inclusion of topics). • Any materials that you have sourced from AI should be rewritten or reconfigured and integrated into your own work and referenced appropriately. It is recommended that this is confirmed by a relevant academic source. • Any support gained from AI should be acknowledged in a statement at the end of the assignment, making clear what the support was, and how you used it and developed it for your own work. Example statements are available in the Student Code of Conduct.	☑
AI EMBEDDED • Use of AI is an integral and expected part of the assessment. • The explicit inclusion of AI within the assessment means that instructions on the expected use will be part of the assessment brief. • Your assessment brief will describe how you should acknowledge the way in which you used AI tools.	

Submission Details

Submission Deadline:	This will be provided on the Moodle submission point.	**Estimated Feedback Return Date**	This will normally be 20 working days after initial submission.
Submission Time:	By 4.00pm on the deadline day.		

Moodle/Turnitin: Any assessments submitted after the deadline will not be marked and will be recorded as a non-attempt unless you have had an extension request agreed or have approved mitigating circumstances. See the School Moodle pages for more information on extensions and mitigating circumstances.

File Format: There will be one submission point on Moodle, which should include the following: Presentation (PRES1) (which you will develop).

Your assessment should be titled with your:

student ID number, module code and assessment ID,
e.g. st12345678 CMP7005 PRES1

Late Submission Window Eligibility Where submissions are eligible for the late-submission window this will be communicated in the relevant assessment submission point within Moodle.

Feedback Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the guidance on how to access your feedback.

All marks are provisional and are subject to quality assurance processes and confirmation at the programme Examination Board.

Assessment Criteria

Learning outcomes assessed

[LO1] Develop software through programming using a suitable high-level general-purpose programming language.
[LO2] Manage file I/O handling in order to access and modify datasets for further analysis.
[LO3] Critically evaluate commonly used libraries used for data analysis tasks such as machine learning and visualisation.
[LO4] Critically evaluate, develop, and implement solutions for processing datasets and solving complex problems in various environments using relevant programming paradigms.
[LO5] Manage version control systems and collaborative programming and project management tools, and their application to real-world problems.

Assessment criteria:

Presentation:	
Motivation for Investigating the Dataset	**5**
Data handling (LO2): Clarity in describing the data import process, including any challenges faced and how they were addressed.	**5**
Critical Evaluation (LO2, LO3, LO4): Critically assess the methodologies (e.g., data cleaning techniques, EDA approaches, machine learning models) you used for your analysis. How do these choices contribute to your understanding of the data and the domain?	**15**
Key Findings (LO3, LO4): In your .ipynb file, present the analysis results clearly using appropriate visualizations, and discuss the key insights and patterns observed to effectively communicate the significance of your findings.	**25**
Application Development (LO1): Demonstrate the working software or platform you developed as part of the project. Show how it integrates with your project**.**	**15**
Critical Reflection (LO3, LO4):	**15**
Provide actionable recommendations based on your findings and the development of your software platform (LO4)	5
Discuss any challenges or limitations you encountered (in terms of data quality issues or model constraints, application development) (LO3, LO4)	5
Identify areas for improvement and suggest how these could be addressed in the future. (LO4)	5
Version Control (LO5): Reflection on the use of version control (How did it help in managing and organizing the work)	**10**
Presentation Style	**10**
Total	**100**

Other skills/attributes developed

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

Ethical	Students will be encouraged to consider the ethical implications of data handling and analysis and will be made aware of the university and/ or other global ethics processes.
Digital	Develop strong digital skills by focusing on programming, software development, and the use of industry-standard tools relevant to the industry, enhancing their digital competence.
Global	Through international research, literature, and real-world case studies, students will be able to build a strong awareness/ appreciation of the importance of being able to think outside the box and find solutions.
Entrepreneurial	Foster an entrepreneurial mindset by requiring students to work on real-world problems. Through collaborative programming and project management, students will learn

	to take initiative, be innovative, and apply professional practices in developing data-driven solutions that could have commercial applications.

Marking/Assessment Criteria

Detailed criteria can be seen on page 8.

Marking/Assessment Criteria

Tasks	1-19%	20%-39% (Clear Fail)	40%-49% (Narrow Fail)	50%-59% (Pass)	60%- 69% (Merit)	70%-79% (Distinction)	80%-100% (Distinction+)
Motivation for Investigating the Dataset	Provides no meaningful motivation for analysing the dataset. Does not identify any relevant aspects of the data or domain. No clear aims, outcomes, or insights are stated or implied.	Minimal or vague motivation. Limited reference to the dataset or domain, with unclear justification. Intended outcomes or insights are missing or poorly articulated.	Some attempt to explain motivation, but weak or superficial. Identifies a few aspects of the data but with limited justification. Outcomes or desired insights are partially stated but lack clarity or relevance.	Provides a basic motivation for conducting the analysis with some relevance to the dataset. Identifies aspects of the data worth exploring and outlines general aims or expected insights.	Clearly explains the motivation behind the analysis and why the chosen aspects of the dataset are important. States coherent aims and possible insights the analysis may reveal.	Offers a strong and well-justified motivation. Demonstrates a good understanding of which aspects of the domain are significant and why. Articulates insightful and relevant expected outcomes.	Provides an excellent, comprehensive, and well-articulated motivation. Clearly identifies the most meaningful aspects of the dataset or domain and justifies their importance. Presents precise, insightful, and well-aligned expected outcomes and deeper insights that demonstrate high-level critical thinking.
Data Handling	Inadequate or unclear explanation of the data handling process.	Little to no clear explanation of the data handling process.	Poor to Limited explanation of the data handling process.	Basic explanation of the data handling process.	Good explanation of the data handling process.	Very Good explanation of the data handling process.	Thorough explanation of data handling process.
Critical Evaluation of Methodologies	No assessment of methodologies. No understanding of choices.	Little evidence of assessing methodologies. Very limited understanding of choices made.	Limited assessment of methodologies, with minimal understanding.	Basic assessment of methodologies with some understanding.	Good assessment of methodologies, demonstrating adequate understanding of their impact.	Very good assessment of methodologies, demonstrating strong understanding and justification of choices.	Comprehensive and critical evaluation of methodologies, demonstrating in-depth understanding and insightful justification of choices.
Key Findings	Insights missing. No visualizations provided.	Minimal discussion of insights with unclear or irrelevant visualizations.	Limited discussion of insights with weak or confusing visualizations.	Basic discussion of insights, supported by some relevant visualizations.	Good discussion of insights, supported by clear and informative visualizations.	Very good discussion of insights, clearly communicated with strong and effective visualizations.	Excellent discussion of insights, with highly effective visualizations and comprehensive communication of key findings.

Application Development	No evidence of a working application or platform.	Minimal evidence of application, poorly integrated with project.	Limited application development with weak or unclear integration.	Basic application demonstrated, with some integration with the project.	Good application demonstrated, with reasonable integration into the project.	Very good application development, well integrated into the project.	Excellent application development, seamlessly integrated into the project and demonstrating innovation.
Critical Reflection	No reflection provided. No discussion of challenges or improvements.	Minimal reflection with little consideration of challenges or improvements.	Limited reflection on challenges and improvements, unclear or superficial.	Basic reflection on challenges, with some suggestions for improvement.	Good reflection with thoughtful discussion of challenges and actionable improvements.	Very good reflection, with insightful discussion of challenges, limitations, and clear suggestions for future improvement.	Excellent critical reflection, with comprehensive discussion of challenges and limitations, and creative, meaningful suggestions for improvement.
Version Control	No evidence of version control use.	Minimal or unclear evidence of version control.	Limited use of version control, poorly organized.	Basic use of version control with some organization.	Good use of version control, reasonably well organized.	Very good use of version control, well documented and clearly organized.	Excellent use of version control, highly organized, consistent, and fully integrated into project management.
Presentation Style	The presentation is unclear, highly disorganized, unprofessional, and difficult to follow. Delivery is unengaging and unprofessional.	The presentation is unclear, disorganized, or unprofessional. The delivery style is very casual and needs significant improvement	The presentation is unclear and disorganized. The delivery style is casual and needs significant improvement.	The presentation is basic, with some clarity issues. The delivery style is fair and needs significant improvement	Presentation is clear, with a few issues. The delivery style of presentation is good but needs some improvement.	Presentation is clear and professional. The delivery style of the presentation is good.	Presentation is exceptionally clear, professional, and engaging. The delivery style of the presentation is very professional.

Further Information

Who can answer questions about my assessment?

Questions about the assessment should be directed to the staff member who has set the task/assessment brief. This will usually be the Module Leader. They will be happy to answer any queries you have.

Staff members can often provide feedback on an assignment plan but cannot review any drafts of your work prior to submission. The only exception to this rule is for Dissertation Supervisors to provide feedback on a draft of your dissertation.

Referencing and independent learning

Please ensure you reference a range of credible sources, with due attention to the academic literature in the area. The time spent on research and reading from good quality sources will be reflected in the quality of your submitted work.

Remember that what you get out of university depends on what you put in. Your teaching sessions typically represent between 10% and 30% of the time you are expected to study for your degree. A 20-credit module represents 200 hours of study time. The rest of your time should be taken up by self-directed study.

Unless stated otherwise you must use the **HARVARD** referencing system. Further guidance on referencing can be found in the Study Smart area on Moodle and using Cite Them Right (use your university login details to access the site). Correct referencing is an easy way to improve your marks and essential in achieving higher grades on most assessments.

Technical submission problems

It is strongly advised that you submit your work at least 24 hours before the deadline to allow time to resolve any last minute problems you might have. If you are having issues with IT or Turnitin you should contact the IT Helpdesk on (+44) 2920 417000. You may require evidence of the Helpdesk call if you are trying to demonstrate that a fault with Moodle or Turnitin was the cause of a late submission.

Late submission and mitigating circumstances

If you are experiencing personal difficulties which are impacting your ability to engage with assessment, there are a range of support options available, including via your Personal Academic Tutor, the Mitigating Circumstances procedure and the Support to Study procedure.

The Mitigating Circumstances policy and procedure sets out when and how students can:

- Submit eligible assessments late but have it recorded as 'on time', up to four times per level of study.
- Self-declare significant exceptional circumstances in order to defer the assessment to the next submission opportunity.

More information about Late Submission and Mitigating Circumstances is available on MetCentral.

Students with a disability or chronic illness should contact the University's Student Wellbeing Team to discuss any appropriate reasonable adjustments which may be made to their learning and assessments.

Academic Misconduct

Cardiff Met takes issues of academic misconduct **extremely seriously.** The University has procedures and penalties for dealing with academic misconduct. These are explained in full in the University's Academic Misconduct regulations and procedures under Volume 1, Section 8 of the Academic Handbook. The Module Leader reserves the right to interview students regarding any aspect of their work submitted for assessment.

Types of Academic Misconduct, include:

Plagiarism, which can be defined as using without acknowledgement another person's words or ideas and submitting them for assessment as though it were one's own work, for instance by copying, translating

from one language to another or unacknowledged paraphrasing. Further examples include:

- Use of any quotation(s) from the published or unpublished work of other persons, whether published in textbooks, articles, the Web, or in any other format, where quotations have not been clearly identified as such by being placed in quotation marks and acknowledged.
- Use of another person's words or ideas that have been slightly changed or paraphrased to make it look different from the original.
- Summarising another person's ideas, judgments, diagrams, figures, or computer programmes without reference to that person in the text and the source in a bibliography/reference list.
- Use of assessment writing services, essay banks and/or any other [illegible] similar agencies (NB. Students are commonly being blackmailed after using essay mills).
- Use of unacknowledged material downloaded from the Internet.
- Re-use of one's own material except as authorised by your degree programme.

Collusion, which can be defined as when work that that has been undertaken with others is submitted and passed off as solely the work of one person. Modules will clearly identify where joint preparation and joint submission are permitted, in all other cases they are not.

Fabrication of data, making false claims to have carried out experiments, observations, interviews or other forms of data collection and analysis, or acting dishonestly in any other way.

How is my work graded?

Assessment grading is subject to thorough quality control processes. You can view a summary of these processes on the Assessment Explained Infographic.

Grading of work at each level of Cardiff Met degree courses is benchmarked against a set of general requirements set out in Volume 1, Section 4 of our Academic Handbook. A simplified version of these Grade Band Descriptors (GBDs) with short videos explaining some of the academic terminology used can be accessed for Foundation, 1st year, 2nd year and 3rd year undergraduate and MSc programmes.

We would strongly recommend looking at the Study Smart